THE FULL RELATIVE PROJ FUNCTOR F IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. Final step 'It is then a simple matter to prove that X represents the functor F' compresses the four-step glue-functors assembly whose complete template is displayed twice in the same chapter (lemma-relative-proj-d, lemma-spec); all four ingredients (subfunctors, representations, open immersions, covering) are established in-span.

Allowed standard prerequisites (editorial).

- Representation of F_d. lemma-relative-proj-d (carded this batch). Each U_d represents F_d with its universal object.
- Affine identification. lemma-relative-proj-affine (same chapter, not carded). Identifies U_{d,V} with an open of Proj of the sections over affine opens.
- Glue-functors template. schemes-lemma-glue-functors (cross-chapter, named; its four-step application is displayed in full in lemma-relative-proj-d and lemma-spec). The assembly pattern repeated by the final step.

Source reference. Original lemma 01NU: sources/stacks-constructions/pinned-original-source.tex, source lines 3868–3868. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01NN: sources/stacks-constructions/pinned-original-source.tex, source lines 3580–3580. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 01NN. The functor F above is representable by a scheme.*

Proof. Let $U_d \to S$ be the scheme representing the functor F_d defined above. Let $\mathcal{L}_d$, $\psi^d : \pi_d^*\mathcal{A}^{(d)} \to \bigoplus_{n\geq 0} \mathcal{L}_d^{\otimes n}$ be the universal object. If $d|d'$, then we may consider the quadruple $(d', \pi_d, \mathcal{L}_d^{\otimes d'/d}, \psi^d|_{\mathcal{A}^{(d')}})$ which determines a canonical morphism $U_d \to U_{d'}$ over S. By construction this morphism corresponds to the transformation of functors $F_d \to F_{d'}$ defined above.

For every affine open $\mathrm{Spec}(R) = V \subset S$ setting $A = \Gamma(V, \mathcal{A})$ we have a canonical identification of the base change $U_{d,V}$ with the corresponding open subscheme of $\mathrm{Proj}(A)$, see Lemma 01NU. Moreover, the morphisms $U_{d,V} \to U_{d',V}$ constructed above correspond to the inclusions of opens in $\mathrm{Proj}(A)$. Thus we conclude that $U_d \to U_{d'}$ is an open immersion.

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

This allows us to construct X by glueing the schemes U_d along the open immersions $U_d \to U_{d'}$. Technically, it is convenient to choose a sequence $d_1|d_2|d_3|\ldots$ such that every positive integer divides one of the d_i and to simply take $X = \bigcup U_{d_i}$ using the open immersions above. It is then a simple matter to prove that X represents the functor F. $\square$